State Transit^{n} Table

Present State ($Q_2Q_1Q_0$)	Next State ($x=0$)	Next State ($x=1$)
000 (S_0)	011	001
001 (S_1)	010	000
010 (S_2)	100	000
011 (S_3)	011	001
101 (S_4)	100	010

$X = 0$

	Flip Flop Inputs ($D_2D_1D_0$)	Outputs $f(x=0)$
S_0	011	0
S_1	010	0
S_2	100	1
S_3	011	1
S_4	100	0

$X = 1$

	Flip Flop Inputs	Output ($X=1$)
S_0	001	0
S_1	000	0
S_2	000	0
S_3	001	1
S_4	010	1